

Why This Order? Inner Linkages Across the “Basic Theoretical Framework” Sequence

0. The backbone: one pricing equation

The ordering in Part I is chosen to build a single object step-by-step and then express increasingly useful representations of it. The central object is the *stochastic discount factor* (SDF) m_{t+1} , defined by

$$p_t = \mathbb{E}_t[m_{t+1}x_{t+1}] \quad \Longleftrightarrow \quad 1 = \mathbb{E}_t[m_{t+1}R_{t+1}]. \quad (1)$$

All sections in the sequence are “different routes” to (i) construct m_{t+1} , (ii) characterize what m_{t+1} can be, (iii) connect m_{t+1} to portfolio/factor representations, and (iv) translate m_{t+1} into time-series predictability.

A compact summary of the logic:

Preferences $\Rightarrow m_{t+1} \Rightarrow$ Portfolio geometry $\Rightarrow$ State prices (GE)
 $\Rightarrow$ SDF existence (No-arb) $\Rightarrow$ SDF diagnostics $\Rightarrow$ Factor structure $\Rightarrow$ PV/predictability.

1. Consumption-based theory $\rightarrow$ (first construction of m_{t+1})

Why start here?

Consumption-based theory provides the first *economic microfoundation* for the pricing kernel:

- A representative agent chooses consumption/portfolio to maximize expected utility.
- First-order conditions yield an Euler equation that can be written in the form (1).
- The resulting m_{t+1} is explicit (in simple models):

$$m_{t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)} \quad (\text{baseline form}).$$

What does it feed into next?

Two immediate translations:

1. Risk premia as SDF covariances:

$$\mathbb{E}_t[R_{i,t+1}^e] = -\frac{\text{Cov}_t(m_{t+1}, R_{i,t+1}^e)}{\mathbb{E}_t[m_{t+1}]}.$$

2. Risk-neutral valuation as reweighting by m_{t+1} (used later in GE and derivatives).

2. Consumption-based theory → Static portfolio theory (geometry and special cases)

Why go to mean–variance next?

Static portfolio theory re-expresses pricing implications using *return space geometry*. It is placed early because it builds intuition and a tractable benchmark:

- Mean–variance introduces the efficient frontier and the tangency portfolio.
- Under conditions such as quadratic utility or elliptically distributed returns, portfolio choice reduces to mean–variance, and the SDF is effectively linear in returns.
- CAPM emerges as a *special factor pricing representation* that can later be embedded into the general SDF framework.

Inner linkage (conceptual)

Consumption-based theory gives m_{t+1} from marginal utility. Static theory gives a *linear projection* of the pricing kernel onto the span of returns:

$$m_{t+1} \approx a - b^\top R_{t+1},$$

which anticipates both (i) “unique SDF within payoff space” and (ii) linear factor models.

3. Static portfolio theory → General equilibrium (complete markets)

Why introduce GE after static theory?

General equilibrium (GE) supplies the cleanest environment where:

- all risks can be traded (complete markets),
- equilibrium implies a unique vector of *state prices* (Arrow–Debreu prices),
- the pricing kernel is tightly pinned down by risk sharing and aggregation.

Inner linkage

Static portfolio theory is “returns-first.” GE is “states-first.” GE converts the return-space geometry into a deeper object:

$$\text{efficient portfolios in return space} \iff \text{state prices / pricing kernel in state space.}$$

This also makes the *risk-neutral measure* representation natural:

$$p_t = \frac{1}{R_{f,t}} \mathbb{E}_t^Q[x_{t+1}],$$

where Q is induced by m_{t+1} .

4. GE (complete) → No-arbitrage (incomplete markets)

Why move from complete GE to no-arbitrage?

Complete markets is a strong assumption. No-arbitrage relaxes it and asks: *what remains true when markets are incomplete?*

- GE provides one route to pricing: preferences + completeness $\Rightarrow$ state prices.
- No-arbitrage provides a weaker route: absence of arbitrage $\Rightarrow$ existence of an SDF that prices traded payoffs.
- In incomplete markets, the SDF may not be unique outside the traded payoff space.

Inner linkage

GE answers “*where does m come from?*” under strong assumptions. No-arbitrage answers “*what must exist regardless of preferences?*” under weak assumptions. This shift is essential to connect theory to empirical practice, where completeness is rarely defensible.

5. No-arbitrage → SDF chapter (properties, uniqueness-in-payoff-space, bounds)

Why a separate SDF chapter after no-arbitrage?

No-arbitrage establishes existence (and potential non-uniqueness). The SDF chapter then organizes:

- **Which SDF is relevant in incomplete markets?** Typically, a unique SDF exists *within the payoff space* (projection results).
- **How large/volatile must m_{t+1} be?** Hansen–Jagannathan-type bounds connect data moments (Sharpe ratios) to admissible SDF volatility:

$$\frac{\sigma_t(m_{t+1})}{\mathbb{E}_t[m_{t+1}]} \geq \max_i \frac{|\mathbb{E}_t[R_{i,t+1}^e]|}{\sigma_t(R_{i,t+1}^e)}.$$

- **Conditional pricing:** information sets and conditional moment restrictions are introduced as a bridge to predictability and ICAPM.

Inner linkage

This section is the “diagnostic hub”:

$$(\text{existence of } m) \Rightarrow (\text{constraints on } m) \Rightarrow (\text{structured approximations of } m).$$

6. SDF → Linear factor models (structured approximation and testable restrictions)

Why factor models after SDF diagnostics?

Factor models are best understood as *low-dimensional parameterizations of the SDF*:

$$m_{t+1} = a_t - b_t^\top f_{t+1}.$$

They appear after the SDF chapter because:

- The SDF chapter clarifies that pricing is about m_{t+1} , not about a specific factor list.
- Factor models impose tractable structure on m_{t+1} and generate beta–lambda restrictions.
- This structure makes empirical estimation/testing feasible (GMM, cross-sectional regressions).

Inner linkage

Factor models translate the abstract restriction $\mathbb{E}[mR^e] = 0$ into operational form:

$$\mathbb{E}[R_i^e] = \beta_i^\top \lambda \quad (\text{in appropriate linear settings}),$$

where β_i summarizes exposure to priced risks and λ are prices of risk.

7. Linear factor models $\rightarrow$ Present value decomposition (time-series implications)

Why end Part I with present value decomposition?

Once m_{t+1} is linked to expected returns, attention turns to *time variation* in discount rates:

- If m_{t+1} (or prices of risk λ_t) is state-dependent, expected returns vary over time.
- Present value decomposition (e.g., Campbell–Shiller style) links price-dividend ratios to expected future returns and cash-flow growth.
- Predictive regressions become interpretable as evidence about the dynamics of discount rates (hence about the dynamics of m_{t+1}).

Inner linkage

Present value decomposition is the “time-series lens” on the same SDF logic:

state variation in m_{t+1} $\iff$ state variation in expected returns $\iff$ predictability and valuation ratios.

Hence it is naturally placed after factor models and conditional pricing discussions.

8. One-sentence transitions (ready to paste between sections)

- **CBAPM $\rightarrow$ Static:** pricing restrictions from marginal utility can be re-expressed as portfolio geometry under mean–variance conditions.
- **Static $\rightarrow$ GE:** return-space results correspond to state-price objects that are uniquely pinned down under complete markets.
- **GE $\rightarrow$ No-arb:** removing completeness shifts the focus from equilibrium uniqueness to existence of an SDF implied by no-arbitrage.
- **No-arb $\rightarrow$ SDF:** once existence is established, admissible SDFs are disciplined via uniqueness-in-payoff-space and volatility bounds.

- **SDF** → **Factor**: factor models are parsimonious parameterizations of the SDF that yield testable beta pricing restrictions.
- **Factor** → **PV**: time-varying prices of risk imply time-varying discount rates, which PV decomposition measures through valuation ratios and predictability.

The notes are organized as a single, cumulative construction of asset pricing around the stochastic discount factor (SDF) and its representations: *Consumption-Based Theory* begins from preferences and optimality, derives the Euler equation, and thereby constructs an explicit SDF that prices all assets; *Static Portfolio Theory* then re-expresses the same pricing logic in return-space geometry (mean–variance frontiers and CAPM), highlighting how pricing restrictions can be understood as projections in the span of traded returns; the *General Equilibrium framework (Complete Markets)* shifts from returns to states, showing how the SDF corresponds to Arrow–Debreu state prices and induces risk-neutral valuation, while the *No-Arbitrage framework (Incomplete Markets)* relaxes completeness and preserves the core pricing insight as a robustness result—absence of arbitrage implies existence (often non-uniqueness) of an SDF that prices traded payoffs; the dedicated *SDF* section consolidates these viewpoints by organizing uniqueness within the traded payoff space, conditional pricing, and diagnostic restrictions (e.g., volatility/bound logic) that discipline admissible discount factors; *Linear Factor Models* impose parsimonious parameterizations of the SDF (or its projection) to obtain operational beta–lambda restrictions (APT/CAPM/ICAPM) suitable for estimation and testing; *Present Value Decomposition* then converts state-dependent discounting—time variation in the SDF or prices of risk—into time-series implications for valuation ratios and return predictability by separating cash-flow news from discount-rate news; with this core toolkit in place, *Asset Pricing Puzzles and Suggested Solutions* interprets empirical tensions as failures of baseline SDF dynamics (the dual puzzle) and develops economically motivated “SDF amplifiers” through *Epstein–Zin Preferences*, *Incomplete Markets*, *Rare Events*, and *Habit Persistence*; *Financial Derivatives* applies the same SDF/risk-neutral logic in continuous time, first building the stochastic-calculus machinery in *Brownian Motion and the Stochastic Integral* and then deriving *Option Pricing* (and related option classes) via replication and equivalent-martingale-measure valuation; *Topics in Asset Pricing* extends the framework without changing the pricing spine by enriching solution methods and state dynamics (*Martingale Methods and Continuous-Time Portfolio Choice*, *Learning Under Uncertainty*, *Markovian Asset Pricing and Long-Run Valuation*), incorporating model uncertainty (*Robustness to Misspecification*), broadening economic environments (*Production Economies*, *Demand-Based Asset Pricing*), and developing preference dynamics (*Habit with Recursive Utility*, *Recursive Utility in Continuous Time*); finally, *Reduced Form Empirical Analysis* closes the loop from theory to measurement by laying out empirical approaches, *SDF Extraction* (including characteristic-based dimension reduction), *GMM for Linear Factor Models*, *Dynamic Factor Models*, *Bayesian Learning with Subjective Beliefs* and *Subjective Expectations and Investor Decisions*, and applied domains including *Fixed Income*, *Machine Learning in Asset Pricing*, *High-Dimensional Investor Decisions*, and *International Asset Pricing*; the *Appendix* provides the recurring technical foundations—utility and aggregation results, distribution comparisons, Brownian motion and Itô calculus, change of measure, and mean/variance estimation tools—that support the full arc from microfoundations to no-arbitrage, diagnostics, applications, and empirics.